Regression Exercises 2026

Course: ML + DL
Code: 650.025
Term: Winter 2026

*Questions marked with * are exam level questions (but not necessarily exam length).*

Submission Requirements (moodle):

1. A pdf document/s containing answers to question 1 to 3.
2. A (scaffold/model).py file containing your regression implementation for the practical exercise.
3. All other scripts used - ***Submissions without these/non-modified versions will only be able to score maximum 1 out of 2.***

Question 1

What tasks would you solve with linear regression? When would it not be suitable?

Question 2

You are investigating the relationship between age and the amount of exercise people do. You take a survey of 10 people at a local park and record their age and the amount of hours they exercise per week. The results are as follows:

Age	**Hours of Exercise Per Week**
22	7.8
28	6.5
31	7.1
35	5.9
40	5.2
45	4.6
50	4.9
55	3.4
62	3.0
68	2.1

a. Fit a Linear Regression model to this data **on paper**, where y is the Hours of Exercise and x is the age. You should state your answer in the form y = mx + c.
b. Using this model, predict the expected number of hours of exercise a week an 80 year old may partake in. Why is this not a reasonable prediction to make?
c. Why may this model not hold up in practice?

Question 3*

Let $X \in \mathbb{R}^{n \times (d+1)}$ be the input matrix (n samples, d features + a bias term), $y \in \mathbb{R}^n$ be the output vector and $\theta \in \mathbb{R}^{d+1}$ be the weight vector.

a. Derive the OLS solution that minimises the residual sum of squares $\|y - X\theta\|^2$.
b. Under what conditions does this solution fail?
c. Name one practical fix for the above condition and explain why.

Practical Exercise

In the data there are 7 input variables labelled x_1 to x_7 and a target variable labelled y. Your task is to implement a regression model **using NumPy ONLY** to predict the target variable from the inputs. It should be vectorised (i.e. operate on the entire dataset at once) and can use as many basis functions (or none at all) as you like.

Your class must be named `Regression` and must implement `fit(self, X, y)` and `predict(self, X)`, where `X` and `y` are NumPy arrays.

You are provided with 4 files:

- **data_train.csv**: contains both input and output observations to train and evaluate your proposed regression model with
- **data_analysis.ipynb**: this is a script to get you started with data analysis
- **scaffold.py**: the starter regression model.
- **train.ipynb:** you can train your model with this notebook

You can change any of these files as you wish but you MUST change the model.

TIP: I recommend inspecting the data first to try to find patterns or relationships and also splitting the data into training and validation sets. Good luck!

You can find the files on moodle or via this link:
https://archie-c.github.io/courses/deep-learning-aau/topics/01-regression/exercises/practical-exercise/
Here you can also upload your model file and score it against your classmates' solutions.